

RESPONSE PAPER:

This study presents an innovative approach that merges cloud computing and Natural Language Processing (NLP) technologies to manage and query Building Information Modeling (BIM) data, which expands massively throughout the lifecycle of Architecture, Engineering, and Construction (AEC) projects. The most remarkable feature of the system is that it enables non-expert staff unfamiliar with Industry Foundation Classes (IFC) standards, or mobile device users with limited interactive screen space, to instantly find answers to their questions using everyday language, such as asking for the "quantity of beams of second and third storey". Keywords and constraints extracted from natural language sentences, which represent user requirements, are successfully mapped to IFC entities and properties via the International Framework for Dictionaries (IFD). The retrieved data is then visualized into tables, charts, and animations according to the user's needs, making information access highly practical. A case study conducted on the Kunming Airport terminal containing over 1.1 million objects and up to 50 GB of data clearly demonstrates the speed and industrial viability of the proposed architecture on large datasets.

However, as the authors transparently state, this model has certain technical limitations that need to be addressed. Since the matching success of the system relies heavily on the IFD library's capacity, it is essential to continuously enrich this dictionary with AEC domain concepts. Furthermore, the current NLP module only supports simple sentences and noun phrases; it is not yet capable of processing complex queries involving verbs, complicated calculations, or mathematical operators. Finally, although the use of a distributed database (MongoDB) and the MapReduce framework significantly accelerates data retrieval and pre-join operations, figuring out how to functionally display the generated detailed multi-axis charts and dashboards on small mobile screens remains a challenge left for future research.

Question 1: The current system can only process simple noun phrases and does not support verbs or mathematical operators. What kind of algorithmic improvements should be made to the NLP architecture so that the system can comprehend complex commands such as "calculate the volume difference between zone A and zone B"?

Question 2: What are the structural advantages of choosing a MongoDB (NoSQL) and MapReduce architecture over traditional SQL in terms of overall query speed and "pre-join" operations within interconnected, hierarchical IFC data

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